

PROJECT WORK

Project Title

Retail POS System (Console-Based Application using Core Java)

```
PS E:\Btech\CDAC-JAVA-2w\Retail-POS-System> & 'C:\Program Files\Java\jdk-26\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\offic\AppData\Roaming\Code\User\workspaceStorage\319c212f3d342d25ec7b31e79ec86997\redhat.java\jdt_ws\Retail-POS-System_b804cbda\bin' 'com.retailpos.main.MainApp'
```

```
=====
```

RETAIL

POS SYSTEM v1.0

```
=====
```

Developed by Team of RetailRangers
(Suraj Arya, Aryan Pandey, Vaivabhai Anand, Atul Mishra)

```
=====
```

Fig. 2: Initial Screen of Retail POS System (Console-Based)

Introduction to the Project

During the training, we developed a **Retail Point of Sale (POS) System** using Core Java. It is a **console-based application**, which means it runs on the command prompt and does not use any graphical interface. The system works in a menu-driven way where users can select options and perform different operations easily.

The main purpose of this project is to manage retail shop activities such as product management, billing, customer details, and inventory. Instead of using a database, all the data is stored in simple text files. This helped us understand file handling in Java and how data can be saved and managed without using complex systems.

This project helped us understand how real-world applications are developed using Java. We learned how to apply OOP concepts, file handling, collections, and exception handling in one project. It also improved our coding skills, problem-solving ability, and confidence in developing real applications.

Objective of the Project

The main objectives of this project were:

- To apply **Core Java and OOP concepts** in a real project
- To understand **file handling** for storing data
- To create a simple **billing and inventory system**
- To improve **coding and problem-solving skills**
- To learn how to work in a **team environment**

Team Structure

This project was developed by a team of 4 members, where each member had a specific role and responsibility. The work was divided into different modules so that the project could be completed in an organized and efficient way.

- **Suraj Arya (PRN: STT-25128071920):**
I worked on the **main application (MainApp.java)**, which controls the overall flow of the program. I created the menu-driven interface where users can select different options like product management, billing, and reports. I also handled user login and authentication. My main role was to connect all modules together and ensure the system runs smoothly.
- **Aryan Pandey (PRN: STT-25128071902):**
He worked on the **model classes** such as Product, Customer, and Bill. These classes are used to store and manage data in the program. He implemented encapsulation using private variables and getter/setter methods and designed the data structure of the system.
- **Vaibhavi Anand (PRN: STT-25128071922):**
She handled the **business logic part** of the project, including inventory management and billing system. She implemented features like adding/updating products, managing stock, and calculating bill amounts with GST.
- **Atul Anand Mishra (PRN: STT-25128071903):**
He worked on **file handling and exception handling**. He managed reading and writing data from text files and created custom exceptions to handle errors like invalid input and insufficient stock.

Working in a team helped us understand how real software projects are developed in parts. Each member focused on their task, and finally all modules were combined to build the complete system. This improved our teamwork, coordination, and practical development skills.

Project Structure

The project follows a proper modular structure where different functionalities are divided into packages. This makes the system easy to understand, manage, and maintain. Each package has a specific role, and all parts are connected together to form the complete application.

Directory Structure:

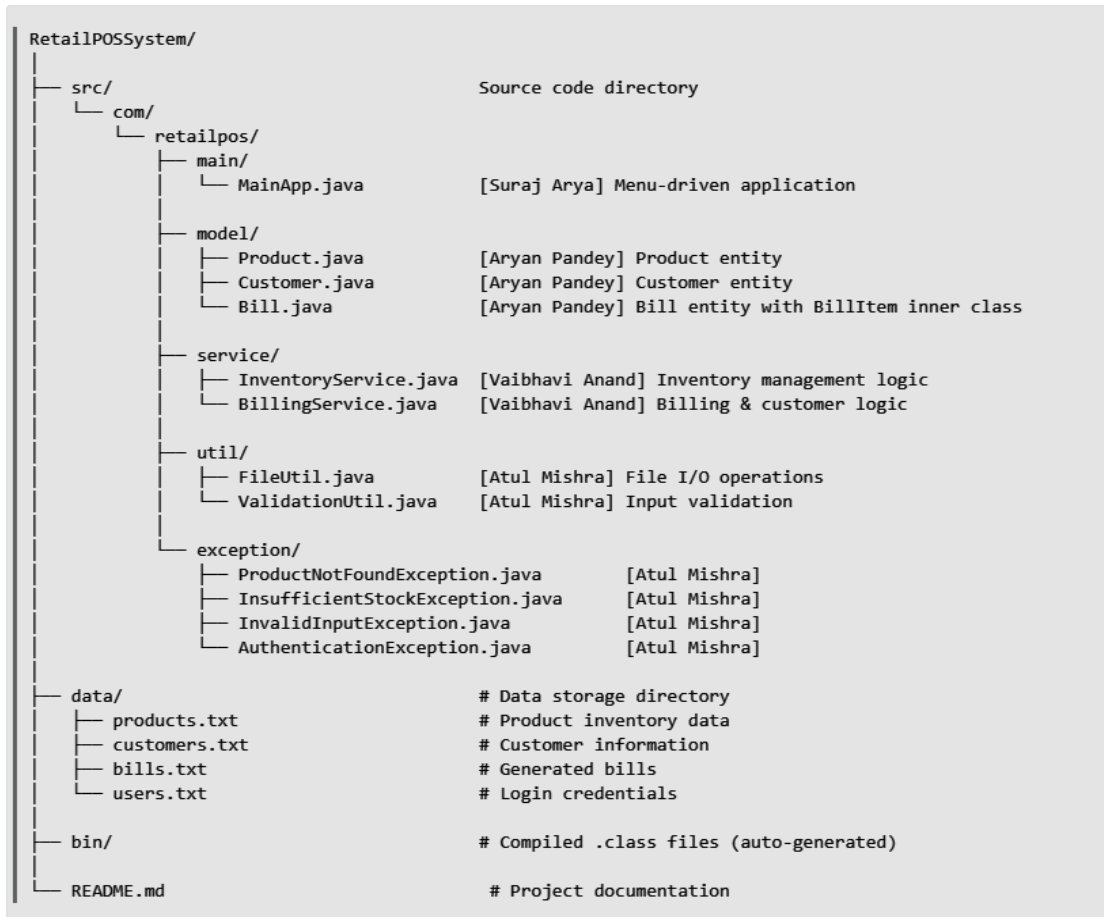


Fig. 3: Directory Structure

Method Call Flow:



Fig. 4: Method Call Flow

Technologies Used

- Core Java
- Object-Oriented Programming (OOP)
- File Handling (Text Files)
- Collections Framework (ArrayList, HashMap)
- Exception Handling
- IDE (Eclipse / IntelliJ IDEA)

System Features

The system provides different features that help in managing retail shop operations in an easy and efficient way. These features make the application useful and user-friendly.

✓ User Login System

- Login using username and password
- Different roles like Admin and Cashier

```
USER LOGIN

Username: admin
Password: admin123

? Login successful!
Welcome, admin (ADMIN)

MAIN MENU

1. Product Management
2. Inventory Management
3. Billing System
4. Customer Management
5. Reports
6. Exit

Enter your choice: 1
```

Fig. 5: User Login

✓ Product Management

- Add, update, delete products
- Search and view product details

PRODUCT MANAGEMENT

1. Add New Product
2. View All Products
3. Search Product
4. Update Product
5. Delete Product
6. Back to Main Menu

Enter your choice: 2

INVENTORY LIST

ID	Product Name	Category	Price	Stock
P005	Headphones	Electronics	Rs.2500.00	25
P004	Monitor	Electronics	Rs.8500.00	20
P007	USB Cable	Accessories	Rs.150.00	100
P006	Printer	Electronics	Rs.12000.00	10
P001	Laptop	Electronics	Rs.45000.00	14
P003	Keyboard	Electronics	Rs.1200.00	30
P002	Mouse	Electronics	Rs.500.00	49
P010	Backpack	Accessories	Rs.1500.00	35
P009	Charger	Accessories	Rs.800.00	45
P008	Phone Cover	Accessories	Rs.300.00	80

Total Products: 10

Fig. 6: Product Management

✓ Inventory Management

- Check stock
- Update stock
- Low stock alert

INVENTORY MANAGEMENT

1. Check Stock
2. Add Stock
3. Low Stock Alert
4. Inventory Value
5. Back to Main Menu

Fig. 7: Inventory Management

✓ Billing System

- Create bills
- Add multiple products
- Calculate total amount with GST (18%)

BILLING SYSTEM

1. Create New Bill
2. View All Bills
3. Search Bill
4. Back to Main Menu

Enter your choice: 2

BILLS HISTORY

Bill ID	Cust ID	Customer Name	Date	Amount
B0001	C001	Aryan Pandey	01-01-2026 23:23:49	Rs.53690.00

Total Bills: 1

Fig. 8: Billing System

✓ Customer Management

- Add and view customers
- Track customer purchases

CUSTOMER MANAGEMENT

1. Add New Customer
2. View All Customers
3. Search Customer
4. Back to Main Menu

Enter your choice: 2

CUSTOMER LIST

ID	Name	Phone	Email	Total Spent
C002	Rahul Sharma	9876543210	rahul.sharma@email.com	Rs.0.00
C001	Aryan Pandey	7410741010	Aryan@gmail.com	Rs.53690.00
C006	Amit Kumar	9876543212	amit.kumar@email.com	Rs.0.00
C005	Vijay Reddy	9876543214	vijay.reddy@email.com	Rs.0.00
C004	Sneha Patel	9876543213	sneha.patel@email.com	Rs.0.00
C003	Priya Singh	9876543211	priya.singh@email.com	Rs.0.00

Total Customers: 6

Fig.9: Customer Management

✓ Reports

- Sales report
- Inventory report
- Customer report

The screenshot shows a Java application window titled 'REPORTS'. Inside, there is a menu with four options: 1. Sales Report, 2. Inventory Report, 3. Customer Report, and 4. Back to Main Menu. Below the menu, the user has entered '1' as their choice. The application then displays the 'SALES REPORT' with the following data:

```

=====
                        SALES REPORT
=====
Total Bills Generated      : 1
Total Items Sold          : 2
Total Sales (incl GST)    : Rs. 53690.00
Total GST Collected      : Rs. 8190.00
Total Sales (excl GST)    : Rs. 45500.00
Average Bill Amount       : Rs. 53690.00
=====
  
```

Fig. 10: Reports

File Handling System

Instead of using a database, all data is stored in **text files**, such as:

- products.txt → Product details
- customers.txt → Customer details
- bills.txt → Billing records
- users.txt → Login credentials

This helped us understand how file handling works in Java.

Working of the Project

1. User logs into the system
2. Selects an option from the menu
3. Performs operations like adding products or creating bills
4. Data is stored or updated in text files
5. Output is displayed on the console